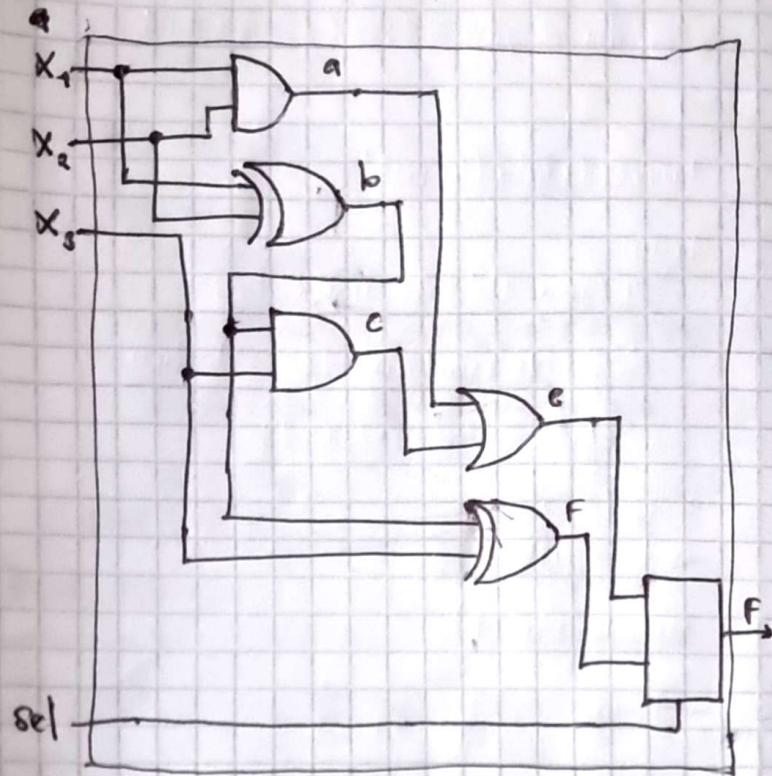


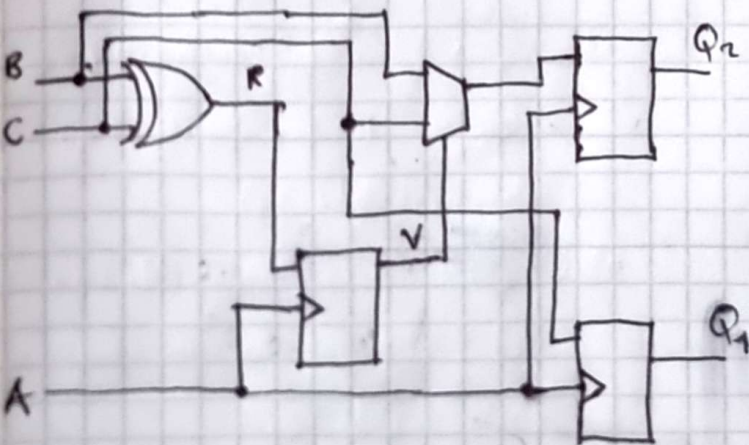
TD2 (3)

Ex 1



Additionner Complément avec sélection entre le retenue et la somme

EXE 2

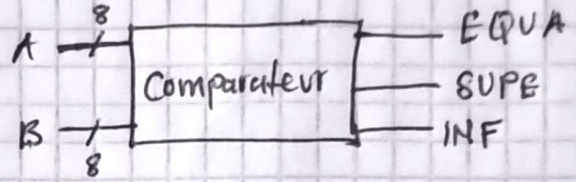


EX 3

$EQUA \leftarrow '1'$ when $A = B$ else $'0'$;

$SUPE \leftarrow '1'$ when $A > B$ else $'0'$;

$INF \leftarrow '1'$ when $A < B$ else $'0'$;



EXE 4

DIN	OBP	EVP
0 0 0 0	0	1
0 0 0 1	1	0
0 0 1 0	1	0
0 0 1 1	0	1
0 1 0 0	1	0
0 1 0 1	0	1
0 1 1 0	0	1
0 1 1 1	1	0
1 0 0 0	1	0
1 0 0 1	0	1
1 0 1 0	0	1
1 0 1 1	1	0
1 1 0 0	0	1
1 1 0 1	1	0
1 1 1 0	1	0
1 1 1 1	0	1

```

library ieee;
use ieee.std_logic_1164.all;
entity Parite is
    port (
        BIN: in std_logic_vector(3 downto 0);
        OBP: out std_logic;
        EVP: out std_logic;
    );
end parite;

```

Architecture arch OF parite is

```

    signal x: std_logic;
begin
    with BIN select
        x <= '1' when "0000" | "0011" | "0101" |
            '0' when others;
    EVP <= x;
    OBP <= not(x);
end arch;

```

Architecture arch OF parking is

```

    signal x: std_logic_vector(3 downto 0);
begin

```

```

    process (clk, rst)
        if (rst = '1') then x <= "0000";
        elsif (clk'event and clk = '1') then
            if (UP = '1') then x <= x + 1;
            elsif (Down = '1') then x <= x - 1;
            end if;
        end if;
    end process;

```

```

    PC <= '1' when x = Max else '0';
    PB <= '1' when x <= Max else '0';

```

EXER 5

```

library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;
entity Parking is
    port (clk, rst: in std_logic;
        UP, Down: in std_logic;
        Max: in std_logic_vector(3 downto 0);
        PB, PC: out std_logic);
end Parking;

```